

Learning Physics, Not Energy: A Multimodal Benchmark for Neutrino Detectors

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Abstract

The experimental search for neutrinoless double-beta decay (NLDBD) would answer one of the most important questions in physics: Why is there more matter than antimatter in our universe? A key challenge in NLDBD searches is identifying NLDBD signal events from many unwanted background events. Deep learning classifiers have shown strong performance on this task, but they often exhibit energy shortcut learning: instead of learning event topology, which more directly reflects the underlying physics, they exploit event energy as an easier predictive feature. In this work, we introduce a comprehensive benchmark for quantitatively understand deep learning classifiers in NLDBD searches. Our benchmark makes three contributions: We coordinated the public release of four NLDBD datasets spanning different modalities: time-series, point-cloud, and sparse 2D image. We then designed two metrics to quantitatively understand topology-based classification power and the degree of energy shortcut learning. Based on datasets and metrics, we conducted a comprehensive benchmarking study over a set of modality-specific deep learning models as well as a unified Transformer model across applied to all modalities. Our results provide insights into when strong classification performance reflects true physics learning, while establishing a pathway toward foundation models for NLDBD data.

1 Dataset Description

The author team coordinated the release of four public, AI-ready datasets, each derived from a neutrino detector capable of searching for NLDBD. Each data point corresponds to a physics **event** recorded by the respective detector; we therefore use the terms “event” and “data point” interchangeably throughout this paper. Because these detectors rely on different detection technologies, their event span substantially different modalities. Our central benchmarking task is to distinguish signal-like events from background-like events based on event topology, even though the topology manifests differently across data modalities.

As shown in Figure 1, the relevant topological signatures appear differently across detectors and data modalities. The Majorana Demonstrator Arnquist et al. (2023a) is a high-purity germanium detector experiment. Event in the **MJD** dataset can be represented as 1D waveforms with 3,800 time samples. Event topology information is primarily encoded in the waveform’s rising edge: signal-like events tend to exhibit a clean, single-step rise, whereas background-like events more often contain multi-step structure. NEXT Novella et al. (2023) is a gaseous-xenon time projection chamber (TPC) experiment. Event in the **NEXT** dataset Mistry (2026) can be represented as sparse 3D point clouds, where each point represents a localized energy deposition. Here, topology is encoded geometrically: signal-like events typically form a continuous track with high-energy clusters at both ends, while background-like events more often exhibit only one such cluster. The SuperNEMO Petro et al. (2025) experiment is a tracker-calorimeter experiment. Each event in the **SuperNEMO** dataset Macko (2026b) can be represented as a similar point-cloud structure compare to NEXT, but the points tends to align more in a straight line (track). Signal-like events will tend to have two tracks starting from the same position, while background-like events tend to exhibit a gap between the two tracks. Finally, EXO-200 Anton et al. (2019) is a liquid-xenon TPC experiment. Each event in the **EXO** dataset can be represented as a sparse 2D images, in which event topology appears as localized “bump” structure. Signal-like events tend to have only one such bump, while background-like events tend to have multiple bumps.

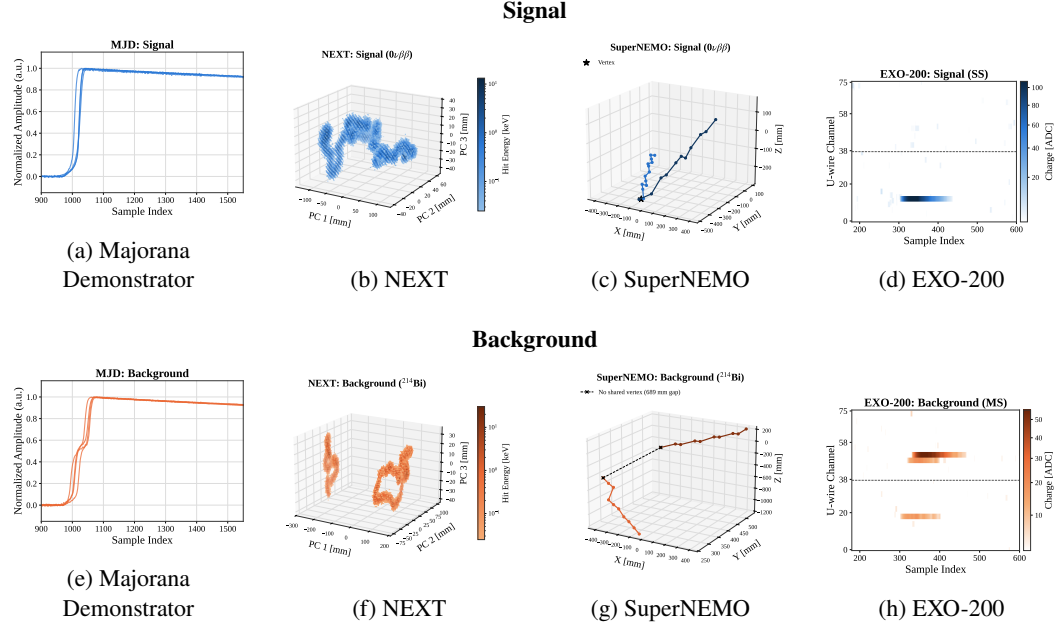


Figure 1: Signal (top) versus background (bottom) examples for the four experiments, each in its native data modality and each illustrating the feature that separates the two classes. Majorana Demonstrator: a PSD-accepted waveform with a single sharp rise (a) versus a PSD-rejected multi-site waveform whose rise shows a visible second step (e). NEXT: a NLDBD 3D hit cloud forming one continuous track (b) versus a ^{214}Bi event whose hits split into two spatially disconnected clusters (f). SuperNEMO: a NLDBD candidate whose two reconstructed electron tracks meet at a common vertex (c) versus a ^{214}Bi candidate whose tracks share no vertex, with a 689 mm gap between their closest points (g). EXO-200: U-wire (charge-collection) channel versus time, where a single-site event produces one collection pulse (d) and a multi-site event produces several separated pulses across both TPC halves (h); the dashed line separates the two halves. Signal is shown in blue and background in orange throughout.

2 Benchmarking Metrics

2.1 Energy Shortcut Learning

When a deep learning classifier is trained on the datasets described in Section 1, it effectively has two choices: it can spend its modeling capacity learning and encoding the complex event topology, or it can resort to a simpler and more obvious feature that already distinguishes signal from background. In NLDBD datasets, energy is exactly such a feature. As shown in Figure 2 (a), the two MJD waveforms correspond to two events with different energies: the taller waveform represents a higher-energy event, while the shorter waveform represents a lower-energy event. This energy difference is one of the most visually prominent features in the MJD dataset, and likewise among the most readily identifiable features across the other datasets.

The problem arises because signal-like and background-like events in the training set often follow different energy distributions. Figure 2 (b) shows the normalized energy distributions of the signal-like and background-like training samples. Without learning the event topology at all, a classifier can exploit this difference with a simple “energy cut”: classifying events above a certain energy as signal-like and events below it as background-like. As shown in Figure 2 (c), this simple “energy cut” approach can already achieve an ROC AUC of 0.9706. In this case, strong classification performance does not necessarily indicate that the model has learned the underlying physics encoded in the event topology; instead, it may simply rely on energy—a simple feature that is often readily apparent by eye—and make its predictions primarily from this shortcut. We refer to this failure mode as energy shortcut learning.

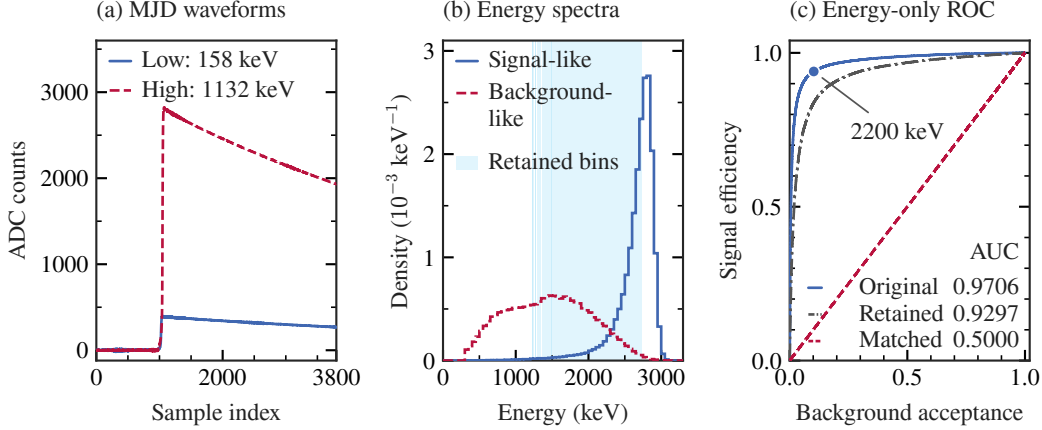


Figure 2: Energy shortcuts in detector data. (a) Baseline-subtracted MJD waveforms from two clean test events with energies of 158 and 1132 keV. (b) Unit-area energy spectra for SuperNEMO $0\nu\beta\beta$ (signal-like) and ^{214}Bi (background-like) test events. Shading marks retained 5 keV matching bins; spectra use 50 keV display bins. (c) ROC curves using energy alone as the score for the original population, retained events with unit weights, and the same retained events after matching. The point marks a 2200 keV threshold.

2.2 Energy-Aware Metric Design

The primary goal of this work is not to eliminate energy shortcut learning, but to provide a systematic benchmark to understand its impact across different NLDBD datasets and modalities. To characterize this effect, we introduce two complementary metrics: a *Energy-matched AUC* that benchmarks the classification power after matching the signal and background energy spectra, and a *Energy Independence Score* that benchmarks stability of the model score distribution with respect to energy within each true class.

For all datasets, we partition events into fixed 5 keV energy bins spanning 0–3000 keV, covering the energy range relevant to the NLDBD searches. An example histogram from the SuperNEMO dataset is shown in Figure 2(b). We refer to an energy bin as *retained* when it contains sufficient statistics for the two benchmarking metrics: Specifically, a bin must contain at least 20 events from each relevant class; for energy-matched AUC, it must additionally lie within the common energy support of the signal and background classes. Bins that do not satisfy these requirements are excluded from the corresponding metric calculation.

Energy-matched AUC ($\text{AUC}_{\text{match}}$) For event x_i , let $y_i \in \{0, 1\}$, E_i , and s_i denote its evaluation label, conditioning energy, and classifier score, respectively, with larger scores favoring $y_i = 1$. Within the common energy support of the two classes, we reweight events so that, in each retained energy bin, the signal and background classes have the same total weight. The target weight assigned to each bin is proportional to the smaller of the two class-normalized bin weights. Letting w_i denote the resulting event weights, normalized to sum to one within each class, the energy-matched AUC is defined as

$$\text{AUC}_{\text{match}} = \sum_{i:y_i=1} \sum_{j:y_j=0} w_i w_j \left[\mathbf{1}(s_i > s_j) + \frac{1}{2} \mathbf{1}(s_i = s_j) \right]. \quad (1)$$

This defines a single pooled weighted AUC, including cross-bin comparisons and half credit for ties (Hanley & McNeil, 1982). Energy matching changes only the evaluation weights, without re-training or random resampling. The metric therefore measures signal-background discrimination after balancing the two classes’ energy distributions. Residual energy differences within individual bins, as well as the exclusion of energies outside the common support, limit this interpretation.

Figure 2(c) compares the ROC curves before and after energy matching for the SuperNEMO signal-like and background-like training samples, using only event energy as the classification score (i.e.,

an “energy cut”). Using all events under their original energy distributions gives an inclusive AUC of 0.9706; We then restrict the evaluation to the retained energy bins used for energy matching, but do not yet apply the matching weights. This lead to an unweighted AUC at 0.9297. Finally, applying the energy-matching weights to these same retained events reduces the AUC to 0.5000, indicating no classification power. Once the signal and background energy distributions are matched and reweighted, energy alone provides no classification power between signal and background.

Energy Independence Score (I) The second metric asks whether the score distribution changes with energy within each true class. For class g , let P_{gk} denote its normalized score histogram in energy bin k , and let P_g denote the corresponding histogram pooled across all retained energy bins. Let π_{gk} denote the fraction of the retained event weight for class g that falls within energy bin k . Using Jensen-Shannon divergence in natural-log units (Lin, 1991),

$$I_g = 1 - \sqrt{\frac{\sum_k \pi_{gk} \text{JS}(P_{gk}, P_g)}{\ln 2}}, \quad I = \frac{\sum_g A_g I_g}{\sum_g A_g}, \quad (2)$$

where A_g is the total original weight, before sparse-bin exclusion, of events in class g with finite scores and energies within the specified energy range. Only classes for which I_g can be estimated are included in the average. This calculation uses the original event weights. Higher $I \in [0, 1]$ indicates greater stability of the classifier score distribution across energy, while lower I indicates stronger energy dependence. Importantly, I measures score stability rather than classification power: a random classifier attains $I = 1$ but an AUC of 0.5000. Energy-matched AUC and I therefore provide complementary diagnostics, with the former measuring discrimination after balancing the class energy distributions and the latter measuring how strongly the classifier score varies with energy within each class.

3 Model Design

We design two complementary modeling approaches for the NLDBD datasets: a *native-representation approach* and a *unified Transformer approach*. The first preserves each dataset as closely as possible to its native representation, applying only minimal preprocessing before benchmarking different model group. Each group provide different inductive biases—for example, RNNs emphasize sequential structure, while GNNs emphasize graph structure. Each dataset is then benchmarked across all architecture families to evaluate how different inductive biases affect different modalities. The second approach instead converts each event into a sequence of tokens through tokenization and positional encoding, after which all tokenized events are processed by a common Transformer backbone. This separates the representation of heterogeneous detector data from the architecture used to model it, allowing us to test whether a unified Transformer can operate effectively across substantially different detector modalities. We benchmark both approaches using $\text{AUC}_{\text{match}}$ and I introduced in the previous section.

3.1 Native Representation Approach

At the beginning of the model-design stage, we defined four model groups relevant to the data modalities encountered in NLDBD datasets. Projection and voxel model group operate on regular spatial grids using 2D or 3D convolutions, sparse convolutions, or projection-patch mixing. Point-cloud model group operate directly on unordered hits using shared point features and local geometric neighborhoods. Graph model group represent events through explicit neighborhood connectivity and message passing over fixed or learned graphs. Finally, sequence, state-space, topology, and hybrid model group capture ordered context, topological summaries, or combinations of multiple representations. We developed 23 classification models across these four groups and benchmarked them on the NEXT dataset, selecting the best-performing model from each group for the subsequent cross-dataset study. Details of this model-selection study are provided in Appendix ???. The four selected models are:

- **Multi-view CNN.** A shared convolutional encoder processes three complementary views, whose embeddings are combined through late fusion. This architecture captures local patterns while preserving view-specific information, and the same backbone can operate on either spatial projections or waveform spectrograms.

- **Static GINE.** Static GINE performs edge-aware message passing over a fixed nearest-neighbor graph (Xu et al., 2019). For spatial data, the graph preserves local neighborhood relationships that may be obscured by projection; for time series data, it is constructed in pulse-shape coordinates.
- **BiGRU.** BiGRU processes two complementary Hilbert orderings using bidirectional recurrence (Cho et al., 2014), providing a sequential baseline for aggregating information across ordered event points.
- **PointMamba-lite.** PointMamba-lite applies input-dependent state-space updates along ordered event points, providing an alternative mechanism for long-range context aggregation (Gu & Dao, 2023; Liang et al., 2024). It is our reduced local implementation and is not intended as a reproduction of the official PointMamba architecture.

Each of these four architectures is subsequently benchmarked on every NLDBD dataset. To accommodate different modalities while preserving the architecture itself, each model uses a detector-specific input adapter together with an architecture-appropriate optimization configuration. Additional details are provided in Appendix E.1.

3.2 Unified Transformer Approach

The central idea of this approach is to use a common Transformer backbone across all detector modalities. To enable this, each event must first be converted into a suitable sequence of tokens, together with positional or structural information added through positional encoding. Our main benchmarking question is therefore how different combinations of tokenization strategy and positional encoding affect different detector modalities. For detector d and tokenization strategy r , the tokenizer maps an event x to

$$\mathcal{T}_{d,r}(x) = (\mathbf{F}, \mathbf{P}, \mathbf{m}),$$

where $\mathbf{F}$ contains the token features, $\mathbf{P}$ contains the positional or structural descriptors supplied to the positional encoder, and $\mathbf{m}$ indicates valid tokens. A learned projection maps the token features into the Transformer embedding space. Separate models are trained for each dataset and representation, while the core Transformer architecture is held fixed across all experiments.

We compare three tokenization strategies that represent detector data at different levels of granularity. *Entity tokens* preserve individual detector measurements: NEXT and SuperNEMO use detector or tracker hits, MJD uses selected waveform samples, and EXO-200 uses sensor-time points balanced across its six sensor blocks. *Region tokens* aggregate nearby measurements while retaining local structure: NEXT and SuperNEMO use occupied three-dimensional regions, MJD uses non-overlapping waveform patches, and EXO-200 uses regions defined jointly over sensor blocks, channel groups, and time intervals. *Summary tokens* apply stronger compression by replacing groups of measurements with descriptive statistics: NEXT and SuperNEMO use balanced Morton-ordered spatial groups, MJD summarizes temporal waveform segments, and EXO-200 summarizes sensor-time regions. Together, these tokenization tests the trade-off between preserving individual measurements, retaining local structure, and compressing detector information into higher-level summaries. Detailed implementations and token budgets are provided in Appendix G.

We additionally compare three positional encoding (PE) mechanisms: learned coordinate-MLP encoding, multiscale Fourier features, and rotary positional encoding (RoPE). Learned coordinate-MLP encoding maps the positional or structural descriptors $\mathbf{P}$ through a small MLP into the embedding space, while multiscale Fourier features encode $\mathbf{P}$ using sinusoidal functions at multiple frequencies to capture structure across different spatial or temporal scales. In both cases, the resulting positional representation is added directly to the projected token features, $\text{Proj}(\mathbf{F})$. RoPE instead uses $\mathbf{P}$ to apply descriptor-dependent rotations to the query and key vectors within self-attention.

Across all tokenization and PE combinations, we use the same Transformer backbone: 64-dimensional embeddings, two Transformer encoder layers, four attention heads, a feed-forward dimension of 256, and masked mean pooling followed by a binary classification head. Although input projections and positional encodings can change the total parameter count, and representation-dependent sequence lengths affect computational cost, the core Transformer architecture remains fixed. This design isolates the effects of tokenization and positional encoding while maintaining a consistent backbone across detector modalities.

4 Experiment Results

Within each dataset, all benchmarked models use the same classification target, train/validation/test partition, and held-out test events. The NEXT and SuperNEMO datasets do not provide predefined train/validation/test splits, so we partition them approximately 80/10/10. EXO-200 and MJD provide separate test sets; we therefore split their provided training data into training and validation subsets using a 90/10 ratio.

The definitions of signal-like and background-like events are dataset-specific. For MJD, the provided binary labels directly identify signal-like and background-like waveforms. For EXO-200, the `n_charge_cluster` variable records the number of reconstructed charge clusters (i.e., “bumps”); we define events with `n_charge_cluster == 1` as signal-like and those with `n_charge_cluster > 1` as background-like. For NEXT, simulated NLDBD events are treated as signal-like and simulated ^{214}Bi events as background-like. For SuperNEMO, simulated two-neutrino double-beta decay (2NDBD) events are treated as signal-like and simulated ^{214}Bi events as background-like.

All unified Transformer models use the same optimization procedure and backbone configuration across tokenization and positional-encoding variants. Models are trained with binary cross-entropy using AdamW, a cosine learning-rate schedule, batch size 64, an initial learning rate of 5×10^{-4} , weight decay 10^{-4} , gradient-norm clipping at 1, and early stopping with patience 5, for at most 50 epochs. Checkpoints are selected using validation inclusive AUC. Native-representation models retain their architecture-specific training configurations, but are evaluated on the same test dataset and with the same metrics. Additional implementation details can be found in Appendix A.3.

4.1 Classification Results

Table 1 reports matched AUC, energy independence I , and trainable parameter counts. Within each dataset, all models use the same metric definitions and evaluation protocol, allowing direct comparison between specialized baselines and tokenized Transformers. Matched AUC measures discrimination after balancing the class energy spectra, while I measures within-class score stability with energy.

Specialized-baseline comparison. Static GINE has the highest matched-AUC point estimate on NEXT, EXO-200, and SuperNEMO among the shared classic models. It also has the highest I on EXO-200, where the recalculated values are 0.949 and 0.801, respectively. BiGRU has the highest matched AUC on MJD, while GINE has the highest I with additional PSD-label supervision. These matched-AUC leaders are unchanged by reevaluation. On SuperNEMO, PointMamba-lite has the highest I and now marginally exceeds BiGRU in matched AUC (0.579492745 versus 0.579485174); this small ordering reversal removes BiGRU from the two-metric Pareto set, leaving GINE and PointMamba-lite. These single-run point estimates do not establish statistical significance. Parameter counts and training protocols also differ, including the additional PSD supervision for MJD GINE.

The finer common energy grid lowers I for the EXO-200 and SuperNEMO classic models relative to their historical quantile-energy estimates; both class-level values decline, as documented in Table 5. This reflects a changed estimator resolution and pooling rule, not a change in the fitted models. All reported retained-campaign classic matched AUCs meet the original count and coverage gates. Coverage, overflow counts, sparse-bin losses, and effective sample sizes are reported in Appendix A.

Among the recorded Transformer results, region tokens provide the strongest matched-AUC configuration on each detector: region-MLP on NEXT, region-Fourier on MJD and EXO-200, and region-RoPE on SuperNEMO. Summary tokens remain close to the leading NEXT representation, suggesting that spatial aggregation can preserve much of the useful event structure. Entity tokens are generally weaker on the dense MJD and EXO-200 waveforms, where selecting isolated measurements can discard contiguous pulse structure retained by regional patches. Across the completed MLP and Fourier comparisons, changing tokenization generally produces a larger performance range than changing positional encoding within one tokenization. This supports the conclusion that defining what constitutes a token is a central modeling decision, although the evidence consists of single-seed results.

¹²¹⁴Bi is a radioactive decay physics event that commonly occur in neutrino detectors.

Table 1: Cross-detector classification results. Higher values are better for energy-matched AUC (AUC_{match}) and energy independence (I). Parameters are rounded counts or ranges in thousands across available dataset-specific implementations; ranges are not uncertainty. Within each model family and dataset, bold marks the highest eligible result where recorded precision resolves the comparison; displayed ties use unrounded values where available. Blue marks the highest AUC and I across both model families in each dataset. Highlighting denotes point estimates, not statistical significance. Results are from single training seeds. A dash denotes an unavailable result. [‡]MJD GINE uses additional PSD-label supervision and is excluded from best-result highlighting. Evaluation details are provided in Appendix A.

Model	#Params (K)	NEXT		MJD		EXO-200		SuperNEMO	
		AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>	AUC _{match}	<i>I</i>
<i>Native Representation Approach</i>									
Multi-view CNN	808	0.977	0.971	0.970	0.760	0.863	0.752	0.586	0.910
Static GINE	480	0.995	0.970	0.969 [‡]	0.797 [‡]	0.949	0.801	0.590	0.913
BiGRU	734	0.979	0.971	0.975	0.775	0.929	0.789	0.579	0.921
PointMamba-lite	317	0.925	0.971	0.960	0.773	0.934	0.786	0.579	0.922
<i>Tokenized Transformers</i>									
Entity + MLP	111	0.975	0.975	0.929	0.812	0.941	0.737	0.591	0.975
Entity + Fourier	113–116	0.963	0.974	0.941	0.802	0.947	0.729	0.589	0.974
Entity + RoPE	106–113	0.497	0.512	0.793	0.749	0.946	0.715	0.551	0.865
Region + MLP	111–121	0.988	0.973	0.973	0.798	0.970	0.801	0.591	0.976
Region + Fourier	113–125	0.979	0.973	0.981	0.788	0.972	0.790	0.590	0.977
Region + RoPE	107–116	0.512	0.483	0.916	0.768	0.971	0.789	0.602	0.953
Summary + MLP	111	0.987	0.971	0.965	0.799	0.947	0.776	0.590	0.975
Summary + Fourier	113–116	0.983	0.973	0.969	0.782	0.960	0.802	0.589	0.977
Summary + RoPE	106–114	0.602	0.532	0.878	0.777	0.955	0.773	0.582	0.956

Effect of positional encoding and energy dependence. Positional-encoding preferences remain detector dependent. Coordinate MLPs perform best across the NEXT tokenizations, whereas Fourier features provide the strongest MLP/Fourier results on MJD and EXO-200. SuperNEMO shows smaller differences among the additive encodings, with its reported region-RoPE variant achieving the highest Transformer matched AUC. NEXT and MJD are more sensitive to RoPE, but for different reasons: NEXT supplies physical spatial coordinates, while MJD supplies time together with waveform level and local change. The comparison therefore varies both coordinate semantics and the encoding mechanism, so it does not isolate a universal positional-encoding effect.

Within model subsets evaluated under a common protocol, the strongest classifier is not always the most energy independent. SuperNEMO further shows that high independence can coexist with weak class separation. Energy-matched AUC and I must therefore be interpreted jointly: the former measures discrimination after energy-spectrum balancing, whereas the latter measures within-class score stability with energy.

5 Conclusion

Figure 3 compares model capacity, energy-matched ranking error, and energy independence for 19 NEXT neural configurations. The complete results are listed in Table 10.

The shared classic-model reevaluation separates energy-matched ranking from within-class score stability. The matched-AUC leaders among the available retained-campaign classic models remain unchanged, while the independence values on EXO-200 and SuperNEMO are lower under the finer common grid. The MJD and EXO-200 reevaluation excludes energies outside 0–3000 keV without changing inclusive AUC. The reevaluated Transformers retain the same matched-AUC leaders, while EXO-200 region-RoPE leaves the Transformer Pareto set after its independence score decreases under the fixed grid. These single-run comparisons describe the retained test populations and do not establish statistical significance or performance under distribution shift.

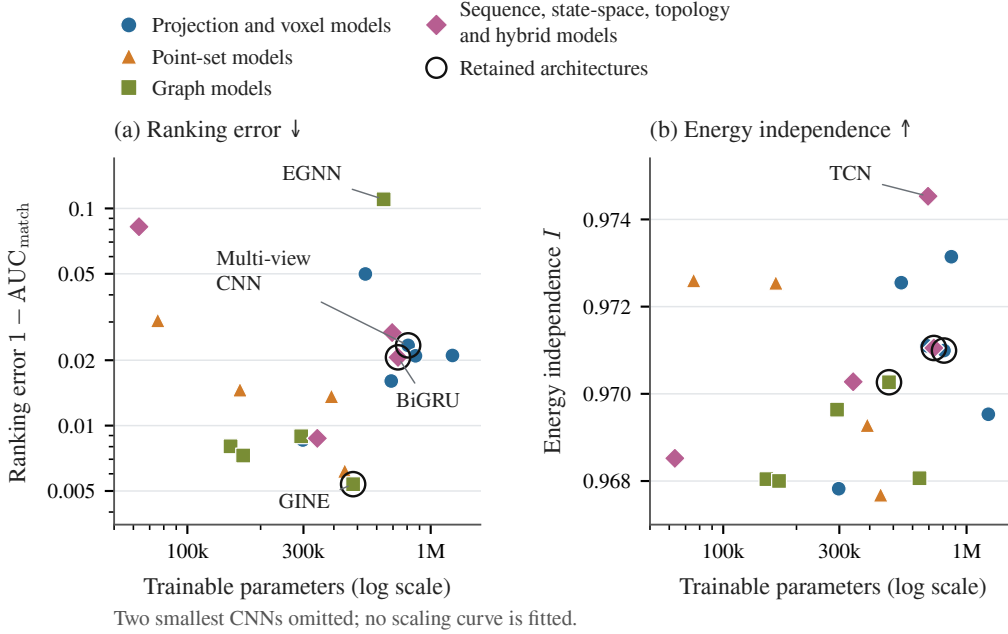


Figure 3: NEXT capacity comparison for 19 neural configurations on 116,549 test events (the shared 5 keV plus overflow protocol), excluding Two-conv CNN and Energy-skip CNN. (a) Weighted pairwise ranking error $1 - \text{AUC}_{\text{match}}$. (b) Energy independence I . Parameter count and ranking error use log scales; I uses a linear scale. Colors and shapes identify four model groups; black rings mark the three retained architectures with campaign results. Earlier exploratory runs are excluded. The full inventory is in Table 10. No scaling curve is fitted.

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References

- G. Anton et al. Search for neutrinoless double- β decay with the complete EXO-200 dataset. *Physical Review Letters*, 123:161802, 2019.
- I. J. Arnuist et al. Final result of the MAJORANA DEMONSTRATOR's search for neutrinoless double- β decay in ^{76}Ge . *Physical Review Letters*, 130:062501, 2023a.
- I. J. Arnuist et al. Majorana Demonstrator data release for AI/ML applications, 2023b. URL <https://arxiv.org/abs/2308.10856>.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pp. 1724–1734, 2014. doi: 10.3115/v1/D14-1179. URL <https://aclanthology.org/D14-1179/>.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023. URL <https://arxiv.org/abs/2312.00752>.
- James A. Hanley and Barbara J. McNeil. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1):29–36, 1982. doi: 10.1148/radiology.143.1.7063747. URL https://www.med.mcgill.ca/epidemiology/hanley/software/hanley_mcneil_radiology_82.pdf.
- Dingkang Liang, Xin Zhou, Wei Xu, Xingkui Zhu, Zhikang Zou, Xiaoqing Ye, Xiao Tan, and Xiang Bai. Pointmamba: A simple state space model for point cloud analysis. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/395371f778ebd4854b88521100af30ad-Abstract-Conference.html.
- Jianhua Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991. doi: 10.1109/18.61115. URL <https://ieeexplore.ieee.org/document/61115/>.
- Miroslav Macko. SuperNEMO simulation dataset for $0\nu\beta\beta$ AI summer school at University of California, Irvine 2026, June 2026a. Dataset release documentation, 18 June 2026, on behalf of the SuperNEMO Collaboration.

Miroslav Macko. SuperNEMO Dataset for 0vbb AI Summer School - UC Irvine, USA (20-21 June 2026), 2026b. [Dataset]. <https://doi.org/10.5281/zenodo.20698789>.

Krishan Mistry. NEXT Simulation Dataset for AI Summer School UC Irvine 2026. *arXiv preprint arXiv:2605.07566*, 2026.

P. Novella et al. Demonstration of neutrinoless double beta decay searches in gaseous xenon with NEXT. *Journal of High Energy Physics*, 09:190, 2023.

M. Petro et al. Status of the SuperNEMO experiment and sensitivity estimates to $0\nu\beta\beta$ and beyond. *PoS, MEDEX2025:034*, 2025.

Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? *arXiv preprint arXiv:1810.00826*, 2019. URL <https://arxiv.org/abs/1810.00826>.

A Full Metric Definitions and Evaluation Protocols

The 0–3000 keV protocol below applies to the reevaluated MJD and EXO-200 classic models, nine EXO-200 Transformer variants, and six MJD Transformer variants with MLP or Fourier position encoding. The NEXT and SuperNEMO classic results and the separate SuperNEMO energy-only illustration retain their preceding evaluation profile, specified below. The three MJD RoPE variants and the NEXT and SuperNEMO Transformer entries retain their historical reported values outside this reevaluation.

A.1 Energy-matched AUC

For event x_i , let $y_i \in \{0, 1\}$ denote its evaluation label, E_i its conditioning energy, s_i its scalar classifier score, and $a_i \geq 0$ its base event weight. Scores are oriented so that larger values favor $y_i = 1$; this convention need not coincide with the stored training label (Table 2). The selected evaluations use unit base weights. We first describe the weighted form implemented by the evaluators. Only finite scores with positive base weight contribute to inclusive AUC; energy-dependent metrics additionally require finite physical energy satisfying $0 \leq E_i \leq 3000$ keV. This range restriction precedes support estimation.

The common support $C = [L, U]$ is the intersection of the two classes’ central energy ranges:

$$L = \max_{c \in \{0,1\}} Q_c(\alpha), \quad U = \min_{c \in \{0,1\}} Q_c(1 - \alpha), \quad \alpha = 0.005, \quad (3)$$

where Q_c is the base-weighted empirical energy quantile of finite-score, finite-energy, positive-weight events in class c within 0–3000 keV. No additional matching region of interest is specified in these runs. For energy bins B_k , define the count n_{ck} and base mass m_{ck} of class- c events in $C \cap B_k$, and let $p_{ck} = n_{ck} / \sum_{\ell} n_{c\ell}$. A bin is retained only if it contains at least 20 positive-weight events from each class. Writing $\mathcal{V}$ for these bins, the implemented overlap target and event weights are

$$q_k = \frac{1\{k \in \mathcal{V}\} \min(p_{0k}, p_{1k})}{\sum_{\ell \in \mathcal{V}} \min(p_{0\ell}, p_{1\ell})}, \quad (4)$$

$$w_i = a_i \frac{q_k}{m_{y_i k}} \quad \text{if } E_i \in C \cap B_k, \quad k \in \mathcal{V}; \quad w_i = 0 \text{ otherwise.} \quad (5)$$

Thus each class has unit total matched weight and assigns mass q_k to bin k . The target favors energy regions represented in both classes and is defined by the minimum of their class-normalized bin masses.

All retained events are pooled into one weighted ROC curve. Equivalently, with $H(u) = 1\{u > 0\} + \frac{1}{2}1\{u = 0\}$,

$$\text{AUC}_{\text{match}} = \frac{\sum_{i:y_i=1} \sum_{j:y_j=0} w_i w_j H(s_i - s_j)}{\left(\sum_{i:y_i=1} w_i \right) \left(\sum_{j:y_j=0} w_j \right)}. \quad (6)$$

This is the weighted pairwise ranking interpretation of AUC (Hanley & McNeil, 1982); equal scores receive half credit. The code implements the equivalent weighted pairwise calculation by sorting scores and processing ties together. It does not average within-bin AUCs: pairs from different energy bins also contribute to Equation 6. Inclusive AUC uses the same ranking calculation with base weights and without energy restriction.

Matching is a deterministic evaluation operation on the supplied split; it does not resample events or alter the fitted classifier. For the reported test results, support and weights are estimated on the test events themselves. The formal matched AUC is withheld if either class retains less than 50% of its original finite-score/finite-energy class base mass, or fewer than two bins survive. Exactly constant common energy exempts only the two-bin requirement; the per-class event-count and coverage gates still apply. Empty, single-class, and insufficient-count bins receive zero matched weight.

Both energy metrics in the MJD and EXO-200 reevaluation use the same global grid after conversion to keV:

$$B_k = [5k, 5(k+1)) \text{ keV}, \quad k = 0, \dots, 598, \quad (7)$$

$$B_{599} = [2995, 3000] \text{ keV}. \quad (8)$$

There are exactly 600 energy bins. Exactly 3000 keV belongs to the last bin; energies below zero or above 3000 keV are excluded and reported separately. Energies are never clipped. Matching additionally requires $L \leq E_i \leq U$, so support boundaries may cut through an energy bin.

Matching equalizes bin masses, not continuous energy distributions within bins. Finite width, finite statistics, trimming, and sparse-bin exclusion limit the interpretation to the retained support and resolution. A score that is solely a function of the matched bin index has AUC 1/2, because its two weighted class distributions are identical; a score depending on within-bin energy need not. This is a property of the weighting construction, not an empirical claim about the trained models.

A.2 Energy independence score

Energy matching measures discrimination after balancing class spectra. The complementary independence score asks whether the *distribution of classifier scores* changes with energy within a true class. Conditioning on class avoids interpreting class separation itself as unwanted dependence. The evaluator groups by supplied physical category, falling back to the binary label when category meta-data are absent. Here each reported task has two such groups. This calculation uses the original base weights, without matching or common-support trimming.

For group g , let P_{gk} be the normalized score histogram in an eligible energy bin, P_g the group’s pooled score histogram, and π_{gk} the fraction of retained base weight in that energy bin. Score bins are shared within a group and obtained from up to 20 weighted score-quantile bins. Using natural logarithms, define

$$\text{JS}(P, Q) = \frac{1}{2} \sum_r P_r \log \frac{P_r}{M_r} + \frac{1}{2} \sum_r Q_r \log \frac{Q_r}{M_r}, \quad M_r = \frac{1}{2}(P_r + Q_r), \quad (9)$$

$$D_g = \sum_k \pi_{gk} \text{JS}(P_{gk}, P_g), \quad I_g = \text{clip}_{[0,1]} \left(1 - \sqrt{D_g / \log 2} \right), \quad (10)$$

$$I = \frac{\sum_{g \in \mathcal{G}_*} A_g I_g}{\sum_{g \in \mathcal{G}_*} A_g}. \quad (11)$$

Here r indexes score-histogram bins, zero-probability terms contribute zero, $\mathcal{G}_*$ contains estimable groups, and A_g is their base weight after finite-pair and 0–3000 keV range filtering, but before sparse-bin exclusion. These MJD and EXO-200 evaluations use the retained-event pooling and eligibility rules specified below. The Jensen–Shannon divergence is bounded by $\log 2$ (Lin, 1991), giving $I_g, I \in [0, 1]$ whenever estimable. Larger values indicate less detected dependence at the chosen binning resolution. The report also retains $\min_g I_g$ to expose a poorly behaving group that the weighted mean could obscure.

This is a histogram-based dependence proxy, not Pearson correlation, distance correlation, mutual information, or a test certifying statistical independence. Its ideal value is one when all compared histograms coincide. Constant predictions also attain one while providing AUC 1/2; a single populated

Table 2: Conditioning variables for classic models and the reevaluated EXO-200 and MJD Transformers. The MJD and EXO-200 reevaluation uses 600 fixed 5 keV bins on 0–3000 keV and excludes energies outside this range (v3). NEXT and SuperNEMO classic results retain 600 regular bins plus one overflow bin (v2). The two metrics retain distinct statistical definitions.

Dataset	Evaluation-positive class	Physical conditioning energy
NEXT	$0\nu\beta\beta$	Summed deposited-hit energy; MeV converted to keV
MJD	Clean pulse	Original calibrated float64 energy (keV), recovered from test-event metadata
EXO-200	Stored label 0 ($n_{\text{CCL}} = 1$)	Reconstructed rotated energy (keV); negate background-oriented logit
SuperNEMO	$2\nu\beta\beta$	Reconstructed calorimeter $E_1 + E_2$ (keV)

energy bin makes the score uninformative about energy dependence. Finite samples can produce nonzero histogram divergences even under population independence, and coarse bins can hide dependence. The score depends on energy coverage, score discretization, and group composition, so it should be compared under a common protocol. No threshold or operating-point diagnostic enters Equations 9–11.

Joint interpretation. High matched AUC indicates ranking ability after the specified spectral balancing; high I indicates stable within-class score distributions across the inspected energy bins. Neither implies the other. Inclusive AUC supplies the complementary performance under the original sampled spectra. Its change after matching is descriptive, rather than a causal decomposition of energy use: both the support and the event weights can change. We report the three quantities together.

A.3 Implementation Details

label selection The stored NEXT labels assign 1 to $0\nu\beta\beta$ and 0 to ^{214}Bi , while SuperNEMO assigns 1 to $2\nu\beta\beta$ and 0 to ^{214}Bi . EXO-200 stores label 1 for events with more than one charge cluster and label 0 for single-cluster events; during evaluation, we reverse the label and score orientation so that single-cluster events form the positive class, following Table 2. MJD assigns label 1 only when all four reference pulse-shape-discrimination criteria pass. The evaluation-positive class therefore has a dataset-specific physical interpretation. NEXT uses an approximately 80/10/10 training/validation/test partition based on class-wise event counts, and different portions of a source file may enter different splits. SuperNEMO uses the same approximate proportions while assigning blocks of complete events within each source category. EXO-200 keeps acquisition runs disjoint across splits. MJD preserves its official test files and divides the provided training data 90/10 into training and validation subsets. Each model for a given dataset reuses the same recorded partition.

Evaluation inputs and orientation. The MJD and EXO-200 reevaluation reuses saved test predictions without training or checkpoint substitution. The unchanged NEXT classic results retain stored scalar classifier scores and convert summed deposited-hit energy from MeV to keV. MJD uses calibrated physical energy recovered in official test-shard/row order. Event identities and the saved labels and energies are verified against the original metadata. The classic adapter’s clipping is removed; Transformer energy caches are restored from float32 to the original float64 values. The three classic binary classifiers and all six reevaluated Transformers retain their single clean-oriented scores, while classic GINE retains the recorded aggregation of four PSD outputs (Appendix E). EXO-200’s stored label 0 is positive ($n_{\text{CCL}} = 1$): we set $y = 1 - y_{\text{stored}}$ and negate the stored background-oriented logit for both metrics, including all nine reevaluated Transformers. The unchanged SuperNEMO classic logits favor $2\nu\beta\beta$ and use reconstructed calorimeter $E_1 + E_2$ in keV. The separate $0\nu\beta\beta$ illustration uses $s = E_1 + E_2$ directly. All selected files have unit base weights. No sigmoid is added to scalar logits: reproducing I requires retaining the score representation as well as the ranking direction.

Exact weighted quantiles. The evaluator sorts finite observations $z_{(j)}$ with positive weights $b_{(j)}$ and assigns cumulative center positions

$$t_j = \frac{\sum_{h \leq j} b_{(h)} - b_{(j)}/2}{\sum_h b_{(h)}}. \quad (12)$$

The empirical quantile $Q(u)$ is linearly interpolated between $(t_j, z_{(j)})$, with endpoint values used outside $[t_1, t_n]$. Stable sorting preserves input order for tied observations. These are the quantiles in Equation 3 and in the histogram construction below. Setting the trim fraction to zero would use each class’s minimum and maximum; the reported configurations instead use 0.005.

Units, boundaries, and population. Physical energy is converted to float64 keV before bin assignment. To avoid a unit-dependent one-step floating-point shift at a theoretical edge, an input MeV value exactly equal to a canonical keV edge divided by 1000 maps to that exact edge. Other nearby observations are not snapped by a tolerance. Internal boundaries enter the bin starting at that boundary, with the explicit 3000 keV exception in Equation 8. Original energies are retained in the audit, including MJD’s 67 events above 3000 keV (66 nonclean and one clean), excluded from the two energy metrics rather than clipped into the 2995–3000 keV bin. Inclusive AUC always uses the original finite-score, positive-weight population, regardless of energy.

Matching calculation and diagnostics. Reproduction first retains finite score/energy pairs with $0 \leq E_i \leq 3000$ keV and positive base weight, then estimates C and assigns the shared 600 bins. Event counts and base masses are computed within $C \cap B_k$. The remaining steps reject bins with fewer than 20 events of either class, compute Equations 4–5, and evaluate one pooled weighted ROC. The class fractions p_{ck} are normalized over all common-support bins *before* invalid bins are removed. Sparse bins are not merged. The formal result is unavailable if support is disjoint, statistics are inadequate, or a reporting gate fails. Coverage for class c is

$$\gamma_c = \frac{\sum_{i: y_i=c, w_i>0} a_i}{\sum_{i: y_i=c, s_i, E_i \text{ finite}, a_i>0} a_i}. \quad (13)$$

This is retained *base* mass, not the matched mass, which is normalized to one in each class. Its denominator precedes both lower and upper energy range exclusions; range filtering cannot inflate the reported coverage. Range loss, subsequent support loss and sparse-bin loss are recorded separately. Both $\gamma_c \geq 0.5$ and at least two valid bins are required. Exactly constant common energy exempts only the two-bin requirement; the coverage and per-class event-count gates still apply. A diagnostic weighted AUC may be saved even when the formal reporting gate fails; it is not substituted for a missing formal result in our table. The effective sample size for any weights v_i is $(\sum_i v_i)^2 / \sum_i v_i^2$. Table 3 reports support and coverage for the reference CNN prediction sets.

Table 3: Matching diagnostics for the multi-view CNN prediction sets under their dataset-specific profiles (Table 2). All reevaluated classic models within each displayed test population use identical test event IDs, labels, physical energies, and base weights. Coverage γ_c uses the original finite score/energy class base mass before energy-range exclusion; ESS uses matched weights.

Dataset	Common support (keV)	Bins	γ_1	γ_0	ESS ₁	ESS ₀
NEXT	[2434.84, 2480.35]	11	0.9882	0.9700	55195	42260
MJD	[18.15, 2613.79]	493	0.9874	0.9631	141205	213947
EXO-200	[538.18, 2662.57]	426	0.9457	0.9414	51924	61627
SuperNEMO	[350.21, 2218.25]	374	0.9900	0.8988	312506	207113

No resampling, new model fitting, bootstrap interval, or multiple-seed average is introduced. Support restriction and matching weights can both alter the AUC relative to the inclusive population; their difference is descriptive, not a causal attribution to energy.

Fixed-grid independence score. The reevaluated MJD and EXO-200 records group events by physical category when supplied, otherwise by binary label. Within each group the evaluator removes

Table 4: Disjoint matching losses and the separate independence sparse-bin loss, reported as positive/negative event counts. N_1/N_0 is the original test population. Range loss precedes support estimation, then matching sparse-bin exclusion. For MJD and EXO-200, range loss includes energies below 0 or above 3000 keV; the $E > 3000$ column is a subset of that loss. For retained NEXT and SuperNEMO results, only negative energies are excluded and $E > 3000$ counts are retained overflow diagnostics. The separate I sparse loss is measured after the applicable range filtering. Counts are shared by reevaluated classic models within each displayed test population.

Dataset	N_1/N_0	Range loss	$E > 3000$	I sparse loss	Support loss	Matching sparse loss
NEXT	64879/51670	0/0	0/0	0/0	767/1550	0/0
MJD	148251/241749	1/66	1/66	137/214	1746/7378	121/1481
EXO-200	66062/74321	3/1423	3/1423	152/257	3587/2932	0/0
SuperNEMO	327680/262144	0/0	0/40	333/271	3275/26530	0/0

nonfinite score/energy pairs, energies outside 0–3000 keV, and zero-weight events, retains only energy bins with at least 20 events, and computes weighted score quantiles using the retained events. Duplicate score edges are removed. For nonconstant scores the extreme edges are moved outward by one floating-point step; constant scores receive one interval centered at that value with half-width $\max(|s|10^{-9}, 10^{-12})$. Score histograms use left-closed/right-open bins with the last right edge included. They are normalized probability masses, not densities. No pseudocount is added.

Let R_g be the retained events, R_{gk} those in bin k , and S_{gr} score-histogram interval r . Then

$$P_{gk,r} = \frac{\sum_{i \in R_{gk}} a_i 1\{s_i \in S_{gr}\}}{\sum_{i \in R_{gk}} a_i}, \quad P_{g,r} = \frac{\sum_{i \in R_g} a_i 1\{s_i \in S_{gr}\}}{\sum_{i \in R_g} a_i}, \quad (14)$$

$$\pi_{gk} = \frac{\sum_{i \in R_{gk}} a_i}{\sum_{i \in R_g} a_i}. \quad (15)$$

Equations 9–10 operate on these histograms. The final group weight A_g includes all finite-pair, positive-weight events in that group within 0–3000 keV, including events from rejected sparse bins. Thus the overall score is not an unweighted class mean and not a mean weighted only by retained events. A group with fewer than 20 finite events or no eligible energy bin is excluded; if no group is estimable, the score is missing. One eligible bin is sufficient for this implementation, unlike the usual two-bin reporting gate for matched AUC.

Changes from historical implementations. The historical EXO-200 evaluations, including the nine Transformer variants, used six class-balanced pooled energy-quantile bins for matching, and a separate eight per-category energy-quantile construction for I . Both energy constructions are replaced by Equation 8; the up-to-20 score-quantile bins remain. Their old I implementation could compare a local energy bin containing only two events and pooled all finite events. The common reference instead requires 20 events per group/bin and computes score edges and the pooled histogram only on retained events. Bin weights use retained group mass; final group weights use eligible population mass before sparse exclusion. These choices follow the former NEXT/MJD fixed-grid estimator and preserve the JS-then-square-root and final group-aggregation definitions. The six MJD Transformer caches used 836 fixed 5 keV bins extending to 4180 keV, without clipping. They now use Equation 8 and the same range exclusions as the other reevaluated models. Their original float64 physical energies are restored before binning. These changes can alter both score-histogram populations and support quantiles, even though the historical MJD Transformer bin width was already 5 keV.

Retained NEXT and SuperNEMO profile. The previously reported NEXT and SuperNEMO classic results, including NEXT’s separate exploratory batches, and the SuperNEMO energy-only illustration retain EnergyBench-unified-5keV-overflow-v2.0.0. This profile uses the 600 regular bins in Equation 8 plus $B_{600} = (3000, \infty)$ keV. Negative energies are excluded, while all finite nonnegative energies enter the support-quantile calculation and the pre-sparse group masses A_g . The overflow bin is subject to the same event-count rules as a regular bin. The score-quantile histograms, overlap target, weighted AUC and original finite-pair coverage denominators use the same definitions as above. These retained results and their figures are not recomputed under the MJD/EXO-200 range restriction.

Table 5: Class-level independence diagnostics under the profiles in Table 2. Retained fractions divide eligible-bin base mass by original finite score/energy class base mass, before energy-range exclusion. Score histograms retain up to 20 weighted score-quantile bins. The audit files include all extended NEXT configurations and full precision.

Dataset / model	I_1	I_0	$\min_g I_g$	Bins _{1/0}	Retained _{1/0}
NEXT / Multi-view CNN	0.9763	0.9643	0.9643	11/11	1.0000/1.0000
NEXT / Static GINE	0.9778	0.9609	0.9609	11/11	1.0000/1.0000
NEXT / BiGRU	0.9753	0.9657	0.9657	11/11	1.0000/1.0000
NEXT / PointMamba-lite	0.9776	0.9632	0.9632	11/11	1.0000/1.0000
MJD / Multi-view CNN	0.7325	0.7763	0.7325	496/524	0.9991/0.9988
MJD / Static GINE	0.8057	0.7911	0.7911	496/524	0.9991/0.9988
MJD / BiGRU	0.7791	0.7730	0.7730	496/524	0.9991/0.9988
MJD / PointMamba-lite	0.7519	0.7865	0.7519	496/524	0.9991/0.9988
EXO-200 / Multi-view CNN	0.7315	0.7709	0.7315	438/484	0.9977/0.9774
EXO-200 / Static GINE	0.7925	0.8093	0.7925	438/484	0.9977/0.9774
EXO-200 / BiGRU	0.7739	0.8033	0.7739	438/484	0.9977/0.9774
EXO-200 / PointMamba-lite	0.7854	0.7863	0.7854	438/484	0.9977/0.9774
SuperNEMO / Multi-view CNN	0.9213	0.8962	0.8962	417/514	0.9990/0.9990
SuperNEMO / Static GINE	0.9241	0.8981	0.8981	417/514	0.9990/0.9990
SuperNEMO / BiGRU	0.9292	0.9105	0.9105	417/514	0.9990/0.9990
SuperNEMO / PointMamba-lite	0.9292	0.9119	0.9119	417/514	0.9990/0.9990

Table 6: Diagnostics for the 15 reevaluated MJD and EXO-200 Transformers using 600 fixed 5 keV bins on 0–3000 keV. All match the CNN test identities, class partitions, physical energies and unit weights within their dataset, so common support and matching losses equal Tables 3–4. Retained fractions and coverage use original finite-pair class mass before range exclusion. The minimum group score is the smaller of the two displayed I_g values. All formal matched AUCs pass the reporting gates.

Dataset / Transformer	I_1/I_0	Bins _{$I_1/0$}	Retained _{$I_1/0$}	K_{match}	γ_1/γ_0	ESS _{$I_1/0$}
EXO-200 / entity-mlp	0.6868/0.7828	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / entity-fourier	0.6545/0.7969	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / entity-rope	0.6583/0.7671	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-mlp	0.7893/0.8115	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-fourier	0.7646/0.8131	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / region-rope	0.7502/0.8246	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-mlp	0.7296/0.8183	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-fourier	0.7894/0.8132	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
EXO-200 / summary-rope	0.7713/0.7740	438/484	0.9977/0.9774	426	0.9457/0.9414	51924/61627
MJD / entity-mlp	0.8388/0.7950	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / entity-fourier	0.8196/0.7913	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / region-mlp	0.7935/0.8013	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / region-fourier	0.7869/0.7881	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / summary-mlp	0.8019/0.7973	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947
MJD / summary-fourier	0.7914/0.7759	496/524	0.9991/0.9988	493	0.9874/0.9631	141205/213947

Reproduction and validation. The MJD/EXO-200 configuration is `EnergyBench-unified-5keV-range-v3.0.0`. The preceding v2 configuration remains available for the retained NEXT and SuperNEMO results; each result records its applicable profile and SHA256 fingerprint. The reproducibility bundle contains the corresponding frozen evaluators, event-input manifests and hashes, full-precision metrics, class-level bin counts and losses, and a table generator that reads these results. Numerical checks cover unit equivalence at all 601 finite edges, adjacent floating-point values, exclusion outside 0–3000 keV, constants, ties, missing and sparse inputs, weighted pair enumeration, and replay of the historical fixed-grid estimator on identical supported inputs. Independent recalculation checks the metrics, coverage gates, and original event sources.

Degeneracies and scope. The unified classification entry point rejects labels outside $\{0, 1\}$ on the entire input array. It also rejects negative or nonfinite weights and misaligned arrays; entirely zero weight yields unestimable metrics. Nonfinite scores are excluded from ROC, and missing energy is additionally excluded from energy-dependent statistics. An absent ROC class makes AUC unavailable; I can still be computed for estimable groups. If no category array is supplied, binary labels define the groups; missing entries in a supplied category array are rejected. The dependence routine filters nonfinite pairs; no correlation coefficient enters I . Constant scores produce identical histograms and $I = 1$. Constant conditioning energy likewise supplies no meaningful test of energy dependence.

The score in Equation 10 averages pairwise Jensen–Shannon divergences *before* taking a square root; it is not an average of Jensen–Shannon distances and not the multi-distribution Jensen–Shannon divergence. We use a descriptive benchmark name for this implemented transformation without claiming that it is a standard test or an original statistical construction.

B MJD Waveform Evidence for Energy-Aware Evaluation

Task, population, and selection. The local MJD release contains six test shards with 390,000 events, including 148,251 clean events. Clean means that all four reference PSD labels—low AvsE, high AvsE, DCR, and LQ—equal one; the formal task is clean versus non-clean, not low versus high energy. Calibrated energy is provided by `energy_label` in keV, and raw waveforms contain 3800 ADC samples (Arnquist et al., 2023b). We choose the detector/run pair with the most clean test events, breaking count ties by detector and run identifier. This gives detector 133, run 50928, with 410 events. Selection uses metadata and labels, before waveform inspection.

The lower and upper energy quartile boundaries are 226.871 and 471.248 keV. The corresponding cohorts contain 103 events each, covering 5.7–226.9 and 471.2–2616.4 keV; the middle group contains 204. For the event displays, we choose the event nearest each cohort’s median energy, with event identifier and row index breaking ties. The selected events have energies 157.975 and 1131.515 keV and identifiers 2407207 and 2585508. This rule does not inspect pulse shape or noise.

Actual waveform preprocessing. For raw waveform samples v_{it} , with $t = 0, \dots, 3799$, define

$$b_i = \frac{1}{200} \sum_{t=0}^{199} v_{it}, \quad \alpha_i = \max_t |v_{it} - b_i|, \quad u_{it} = \frac{v_{it} - b_i}{\alpha_i}. \quad (16)$$

The implementation reads float32 samples, computes the baseline mean in float64 and casts it to float32, then subtracts and normalizes in float32. Division is skipped when $\alpha_i = 0$; no selected event has zero amplitude. This is the common waveform preparation before the STFT or point representation, not an assertion that every classifier directly receives the plotted one-dimensional array. The time axis is sample index; no sampling interval is assumed.

We summarize residual baseline fluctuations by

$$r_i = \left(\frac{1}{200} \sum_{t=0}^{199} u_{it}^2 \right)^{1/2}. \quad (17)$$

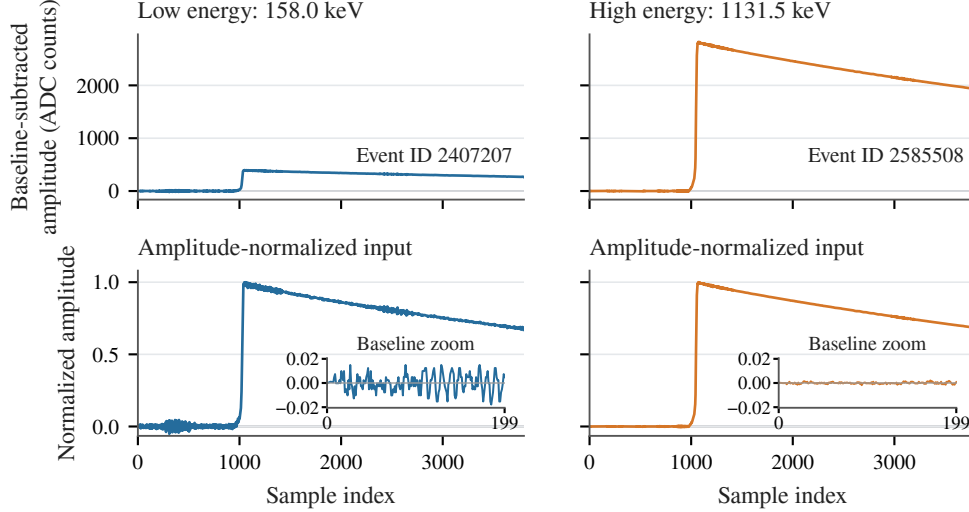
Up to float32 rounding, r_i is baseline RMS in ADC counts divided by α_i . It is a descriptive input statistic, distinct from the benchmark’s energy independence score. The lowest/highest energy cohorts have median r_i values 0.006720 and 0.000889, a ratio of 7.56. Their absolute baseline RMS

medians are much closer, 2.233 and 2.166 ADC counts. Across the 410 events, energy has Spearman correlation 0.993 with α_i , -0.012 with absolute baseline RMS, and -0.826 with r_i . These are single-cohort descriptive statistics, without uncertainty intervals or a claim that they hold for every detector/run. Neither correlation nor these event displays determines the strategy of a trained classifier.

MJD low- and high-energy waveforms

Clean test events; detector 133, run 50928; one event near each cohort median energy.

Low: 5.7–226.9 keV (n=103); high: 471.2–2616.4 keV (n=103)



Subtract the mean of samples 0–199; divide each event by its maximum absolute amplitude. Insets show samples 0–199. Row axes and insets share limits. Sampling interval is not assumed.

Figure 4: Detailed MJD event displays from the fixed detector/run cohort. Top: baseline-subtracted ADC amplitudes. Bottom: normalized waveforms before model-specific representation, with enlarged first-200-sample baselines. The two columns share vertical limits within each row, and both baseline insets use the same scale. Events were selected by energy alone using the stated median rule.

Connection to the saved benchmark. Raw test labels and event IDs match the saved CNN evaluation population. Calibrated energies are recovered in float64 from the original test shards; the old adapter’s float32 conversion and clipping are not used. The 67 events above 3000 keV (66 non-clean and one clean) are excluded from energy evaluation before support estimation; they remain in the original coverage denominator. The common support is $[18.151, 2613.790]$ keV with 493 valid 5 keV bins, retaining 146,383 clean and 232,824 non-clean events. The CNN’s inclusive and matched AUCs are 0.971283 and 0.969851, respectively, while $I = 0.759637$. The small AUC difference does not establish energy-dominated classification. Matching controls the class energy mixture; the independence score asks a separate question about the predictions within each class.

C Energy-Only SuperNEMO Diagnostic

Population and operating point. This diagnostic uses all 196,608 simulated $0\nu\beta\beta$ and 262,144 ^{214}Bi events in the existing shared test partitions. It is separate from the trained $2\nu\beta\beta$ classification task. Energy is the reconstructed calorimeter sum $E_1 + E_2$ in keV, extracted once per event; the identical event-level fields repeated on tracker rows are not counted as separate observations. The released two-electron topology selection applies, with no additional energy cut or random sampling for the

spectra. Class normalization removes differences in simulation sample counts; it does not estimate exposure-normalized physical rates.

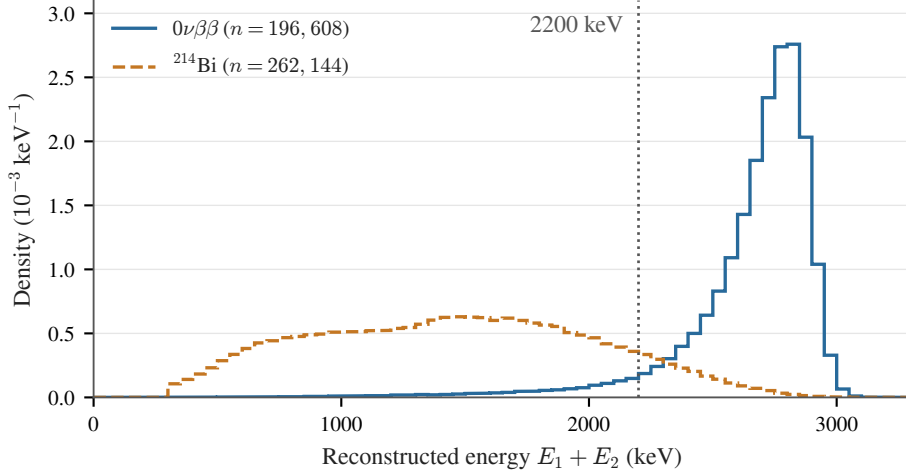


Figure 5: The motivating energy-spectrum difference: 196,608 simulated $0\nu\beta\beta$ and 262,144 ^{214}Bi events from shared SuperNEMO test partitions. The 50 keV display bins have unit area per class, so sample counts do not determine their relative heights. The released two-electron selection applies (Macko, 2026a); no additional energy cut is applied to the plotted spectra. The vertical line marks the illustrative 2200 keV threshold, not a test-optimized operating point.

At the illustrative threshold $t = 2200$ keV, the rule $\hat{y}_i = 1\{E_i \geq t\}$ accepts 184,727 of 196,608 signal events and 26,793 of 262,144 background events. Hence

$$\text{TPR} = 0.939570, \quad \text{FPR} = 0.102207, \quad \text{BA} = \frac{1}{2}[\text{TPR} + (1 - \text{FPR})] = 0.918681, \quad (18)$$

where BA is balanced accuracy, assigning equal importance to both classes. Figure 6 shows the operating-point trade-off. The threshold was specified for illustration rather than selected by maximizing test performance. Neither this cut nor the score $s_i = E_i$ requires training or event topology.

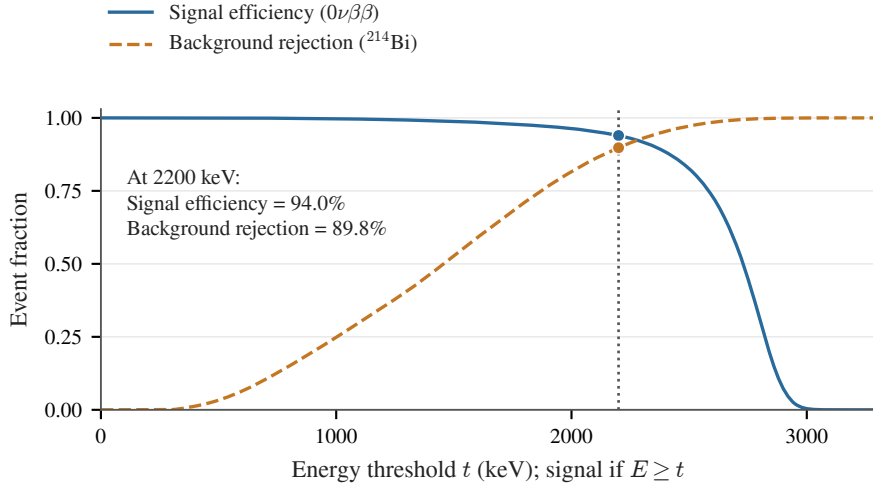


Figure 6: Signal efficiency and background rejection as an energy-only threshold varies in the same SuperNEMO $0\nu\beta\beta$ - ^{214}Bi test partitions. Curves use all events; the display samples thresholds every 10 keV. Markers identify the illustrative 2200 keV cut, with rates calculated directly from event energies.

Matching and coverage. We apply the shared 600 regular 5 keV bins plus one overflow bin to this diagnostic pair, with class-wise 0.5% tail trimming, overlap matching, and at least 20 events of each class per bin. Common support remains $C = [1169.390, 2734.195]$ keV. The slice of the zero-anchored grid intersecting C has edges

$$(1165, 1170, 1175, \dots, 2730, 2735) \text{ keV.} \quad (19)$$

Events must also satisfy $E \in C$, so the boundary bins are partially occupied. Of 314 candidate bins, 283 pass the count requirement; sparse bins are excluded without merging. This retains 99,139 signal and 150,591 background events, or fractions 0.504247 and 0.574459, respectively. Both satisfy the minimum coverage of 0.5, with signal coverage close to the boundary. The matched result therefore describes the retained energy population, not the full signal peak. Reweighting gives effective sample sizes of approximately 34,127 and 64,317. The 50 keV display bins in Figures ??(a,b) and 5 do not set the matching resolution.

The inclusive ROC uses all original energies. The shared energy grid is assigned before support trimming, including 758 signal and 40 background events with $E > 3000$ keV in one overflow bin. All energies are finite and nonnegative. Here C lies below 3000 keV, so overflow events receive zero matched weight; they remain in the coverage denominators and in the separate independence calculation whenever the group/bin count rule passes. No energies are clipped. The trained $2\nu\beta\beta$ task uses the same evaluation rules on its own distinct test population.

Table 7: Energy-only AUC under successive evaluation conditions. Retained event counts are unweighted; only the last row uses matching weights. The populations and weights differ, so these rows are not competing models.

Evaluation condition	Signal events	Background events	AUC
Original test populations	196,608	262,144	0.970613
Common support, unit weights	99,595	172,757	0.934642
Retained 5 keV bins, unit weights	99,139	150,591	0.929715
Retained 5 keV bins, matched weights	99,139	150,591	0.500024

Residual within-bin ordering. For the increasing score $s = E$, let q_k denote the shared matched mass of ordered energy bin k , and let A_k be the energy-only AUC within that bin, with half credit for ties. Cross-bin comparisons give

$$\text{AUC}_{\text{match}}(E) = \sum_{k>\ell} q_k q_\ell + \sum_k q_k^2 A_k = \frac{1}{2} + \sum_k q_k^2 \left(A_k - \frac{1}{2} \right). \quad (20)$$

Equal bin masses balance cross-bin pairs, while within-bin energy differences remain. This identity gives AUC 0.500024 for the fixed 5 keV matching bins; it does not redefine the metric as an average of per-bin AUCs. A score depending only on bin identity would instead have matched AUC exactly 1/2. The independence calculation uses the same energy grid with original base weights and up to 20 score-quantile bins, yielding $I = 0.079210$ for $s = E$. Its class values are $I_{0\nu} = 0.082192$ and $I_{\text{Bi}} = 0.076972$. The 337 and 514 eligible group bins retain 195,169 signal and 261,873 background events, respectively, including their eligible overflow bins. The energy-only score still varies within each 5 keV bin; balanced bin masses therefore do not imply exact continuous-score AUC 1/2. These are deterministic diagnostic values, without bootstrap intervals or multiple-seed averages. They demonstrate the availability of an energy-only decision rule, not the internal strategy of any trained model or a measured failure under distribution shift.

D SuperNEMO Tracker Extent and Energy

For an event with h_i raw tracker-hit positions $\vec{r}_{ij}$ in millimeters, we measure spatial extent by

$$R_{g,i} = \left[\frac{1}{h_i} \sum_j \|\vec{r}_{ij} - \bar{\vec{r}}_i\|^2 \right]^{1/2}, \quad \bar{\vec{r}}_i = \frac{1}{h_i} \sum_j \vec{r}_{ij}. \quad (21)$$

This radius of gyration uses all stored tracker hit centers with equal weights; it is not a fitted track length. Reconstructed energy is the calorimeter sum $E_1 + E_2$, counted once per event. We retain all 262,144 ^{214}Bi and 196,608 $0\nu\beta\beta$ test events, with no missing energies or coordinate summaries.

Figure 7 compares complete populations. Each 100 keV energy column is normalized over all 20 mm extent bins. Columns with fewer than 20 events are masked, affecting one ^{214}Bi and 35 $0\nu\beta\beta$ events. Extents above 800 mm remain in the normalization but are outside the heatmap view; the cumulative plots include their full tails. These display bins are separate from EnergyBench matching bins.

To compare balanced cohorts, events are ranked by energy and then event ordinal within each physical class. We take the first and last quarter: 65,536 events per cohort for ^{214}Bi and 49,152 for $0\nu\beta\beta$. Exact energy ties are split by ordinal, not geometry. The ^{214}Bi quartile boundaries are 1003.0 and 1870.25 keV; the $0\nu\beta\beta$ boundaries are 2583.125 and 2821.5 keV. These relative energy cohorts are not common absolute windows across classes.

Median R_g decreases from 335.97 to 291.14 mm for the illustrative $0\nu\beta\beta$ sample and changes from 281.97 to 294.04 mm for ^{214}Bi . The distributions overlap substantially. Geometry can therefore contain energy-associated information, but these observations do not establish its use by a trained classifier or a universal monotonic size–energy relation. No specific physical or selection mechanism is inferred. The formal SuperNEMO benchmark remains $2\nu\beta\beta$ versus ^{214}Bi .

The current point adapters center events, quantize at 15 mm, and divide coordinates by a fixed 1000 mm scale; CNNs use a fixed 44 mm occupancy grid. They exclude calorimeter energies and do not normalize each event to unit spatial extent. Quantization and aggregation nevertheless change the input, so the raw extent statistic is not a claim about the learned representation.

E Non-Transformer Architectures and Run Protocols

E.1 Evidence Supporting the Four-Model Comparison

Unified reevaluation of earlier NEXT checkpoints provides retrospective context for the representative architectures. In the alternative-model batch, the multi-view CNN has matched AUC 0.955804, compared with 0.947782 for the multiscale CNN and 0.928273 for the dense 3D CNN; Static GINE leads the graph candidates in that batch at 0.981008. In the subsequent candidate batch, BiGRU exceeds the tested dilated TCN (0.933864 versus 0.926085). PointMamba-lite, the only recorded state-space candidate, has matched AUC 0.925386 and $I = 0.971387$; its inclusion broadens the architectural comparison rather than following an overall accuracy ranking.

These are held-out test comparisons on 115,499 events from earlier exploratory batches. They are distinct from the 116,549-event reevaluation used for the available NEXT classic results in Table 1. PointMamba-lite is included in Table 10. The original exploratory batches used different evaluator-code fingerprints; the numbers above now use the same regular-plus-overflow protocol as the current classic results while retaining the earlier checkpoints and test set. Their distinct test population is not pooled with the retained campaign. The four search groups are recorded in the model-family presentation and are reproduced in Table 10. They organize the candidate inventory; the surviving records do not document a prospective validation-based rule selecting one winner from each group. The retained architectures span three of the four groups, including two distinct mechanisms from the sequence/state-space group. The structural rationale and early test comparisons therefore support the comparison set without establishing a recovered formal selection rule.

Parameter-count provenance. Parameter counts in Table 10 are the saved counts of trainable neural parameters, checked against the corresponding run summaries and reevaluated metrics. Neural parameter count is inapplicable to XGBoost. The earlier NEXT PointMamba-lite run summary records 316,993 trainable parameters, consistent with `Benchmarking table.xlsx`; this is displayed as 0.3170 M in Table 10 and 317 K in Table 1. Its retained-campaign predictions remain unavailable. Figure 3 includes only the retained-campaign configurations with available predictions.

Spatial representations. NEXT MV-CNN uses three 128×128 projections with 30 mm bins. For point-based models, hits are centered by the complete-event energy centroid and merged into 15 mm cells. Voxel centers are recentered and divided by 1,000 mm; the two features are the voxel’s fraction

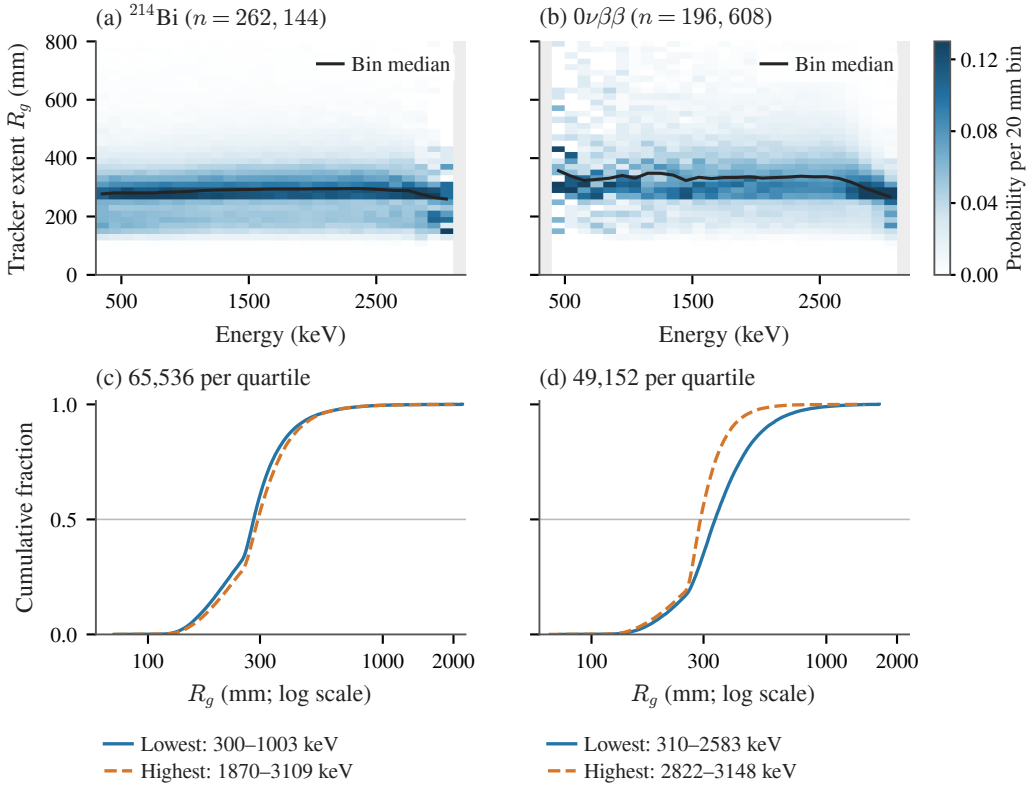


Figure 7: SuperNEMO tracker extent versus reconstructed energy. Top: conditional distributions using 100 keV by 20 mm bins; each energy column is normalized over all extents. Black curves show medians; gray columns contain fewer than 20 events. Extents above 800 mm (0.99% of ^{214}Bi and 1.10% of $0\nu\beta\beta$) are excluded from view only. Bottom: complete cumulative distributions for the lowest and highest energy-ranked quartiles within each class, on a logarithmic extent axis. Cohort sizes and energy ranges are given in the panels.

Table 8: Input adaptations for the four non-Transformer architectures. GINE, BiGRU, and PointMamba-lite share each dataset’s point representation. Model dimensions and representation details are given in Appendix E.

Dataset	MV-CNN input	Graph and sequence input
NEXT	Three normalized spatial energy projections	Centered 15 mm voxels; energy fraction and log hit count
SuperNEMO	Three binary tracker projections	Centered 15 mm tracker cells; occupancy fraction and log hit count
MJD	Three-resolution waveform STFT	512 pooled waveform segments with mean and RMS features
EXO-200	STFT of serialized 226×300 waveform	Same waveform construction after channel-major serialization

of complete-event energy and $\log(1+n)$, where n is its hit count. Events exceeding 512 occupied cells retain the highest-energy cells with deterministic tie breaking, without renormalizing the retained energy fractions. SuperNEMO instead uses tracker coordinates: its CNN projects binary occupancy on 128×128 grids with 44 mm bins. Its point adapter centers the tracker hits geometrically and merges 15 mm cells, replacing the unavailable hit-energy fractions with occupancy fractions. It raises an error above 512 cells instead of truncating. The non-Transformer SuperNEMO inputs exclude E_1 ,

Table 9: Specified common protocol versus recorded training settings for the audited non-Transformer models. Every listed configuration has an upper limit of 50 epochs, weight decay 10^{-4} , gradient-norm limit 1, and seed 42. NEXT settings refer to the three available retained-campaign runs. “Other datasets” means EXO-200, MJD, and SuperNEMO.

Scope	Architecture	Batch	Initial learning rate	Patience
Specified protocol	Neural models	64	5×10^{-4}	5
NEXT recorded	MV-CNN, GINE, BiGRU	64	5×10^{-4}	5
Other datasets	MV-CNN	16	10^{-3}	12
Other datasets	GINE, BiGRU	16	5×10^{-4}	12
Other datasets	PointMamba-lite	4	3×10^{-4}	12

E_2 , and tracker drift radii. These input choices restrict direct energy access but do not establish independence from energy.

Waveform representations. MJD subtracts the mean of the first 200 waveform samples and divides by the maximum absolute event amplitude when nonzero. EXO-200 subtracts a baseline separately for each channel using its first 200 time samples, divides by the event-wide maximum absolute amplitude, and serializes the 226×300 array in channel-major order. For these datasets, MV-CNN uses Hann-window STFTs with FFT sizes (64, 128, 256) and hops (16, 32, 64), followed by a $\log(1 + |\text{STFT}|)$ transform and resizing to 128×128 . The point adapter uses adaptive pooling to form 512 segment means and root-mean-square amplitudes. Its three coordinates are normalized time, amplitude-normalized segment mean, and the first difference of that normalized mean; its two features are the segment mean and RMS. These are pulse-shape coordinates, not calibrated detector positions.

Backbone details. MV-CNN has four residual stages with two blocks each, base width 16, and a 256-dimensional fusion layer. GINE uses hidden width 128, five message-passing layers, $k = 12$ neighbors, and a 160-dimensional classifier. BiGRU uses a 96-dimensional input embedding, two bidirectional layers with hidden size 128, and a 256-dimensional classifier. The two Hilbert orders quantize each event’s point bounding box to 10 bits per axis; the second swaps the first two axes before applying the same ordering. PointMamba-lite uses three blocks with model width 128, inner width 192, state dimension 16, and 32-position scan chunks. Its guarded PyTorch scan is not the official fused Mamba implementation. The backbones use dropout 0.1. MJD GINE has four output logits z_{ij} , one per reference label, and uses

$$q_i = \prod_{j=1}^4 \sigma(z_{ij}), \quad s_i = \log q_i - \log(1 - q_i), \quad (22)$$

where σ is the logistic sigmoid and s_i is its evaluated clean-event score. This product is the implemented score construction; it is not a demonstrated factorization of the labels’ joint conditional distribution. The computation uses summed log-sigmoids and clamps the log probability below zero for numerical stability. EXO-200 uses the same GINE backbone with one output.

Run completion and randomness. The audited non-Transformer classification loaders retain the sampled class proportions; their binary losses do not use class weights, and these paths contain no synthetic data augmentation. NEXT shuffles training slices and buffered events, waveform loaders shuffle training events, and SuperNEMO records proportional sampling without replacement. Input normalization and ordering are dataset-specific preprocessing rather than energy matching of training events.

In the NEXT campaign reported here, MV-CNN, GINE, and BiGRU finish after 19, 36, and 26 epochs, selecting epochs 14, 31, and 21, respectively, by validation inclusive AUC. The loader batch size is 64; the wrapper evaluates these backbones in chunks of 16, concatenates their logits, and applies one batch loss and optimizer step. The corresponding run summaries record Python 3.11.15, PyTorch 2.11.0+cu128, an NVIDIA GeForce RTX 5090, and bfloat16 mixed precision. This environment evidence applies to those runs rather than automatically to every dataset. The audited

non-Transformer training paths add neither class-weighted losses nor data augmentation; the current SuperNEMO stream retains the source class proportions without replacement.

For SuperNEMO, the workbook records 21, 50, 22, and 9 training epochs for MV-CNN, GINE, BiGRU, and PointMamba-lite, with selected epochs 16, 43, 17, and 4, respectively. It records 47 epochs and selected epoch 43 for MJD PointMamba-lite. Earlier logs were incomplete; the current results and epoch counts follow the workbook, whose evaluation status does not establish training completion. The audited non-Transformer configurations set seed 42 and disable enforced deterministic algorithms. Mixed precision is enabled in these configurations except for MJD BiGRU. Seeds therefore do not imply bitwise reproducibility.

Screening scope. The current NEXT registry contains three legacy CNNs and twenty alternative classifiers, including point, graph, sequence, state-space, sparse-convolution, hybrid, mixer, persistence, and boosted-tree approaches. A classification entry need not correspond to a completed retained-campaign run: PointMamba-lite has an earlier exploratory result but remains unavailable in that campaign. The search taxonomy groups the three legacy projection CNNs together with projection, patch-mixer, dense-voxel, and sparse-voxel alternatives. Point-set and graph methods form separate groups; ordered sequence methods, the state-space model, topological classifiers, and the CNN-GNN hybrid form the fourth. This representation-based organization differs from the finer model-family labels in the run registry. Checkpoint selection is documented within completed runs, while a prospective rule for choosing the cross-dataset architectures is not. The boosted-tree classifier in the extended NEXT inventory uses up to 500 estimators, learning rate 0.04, maximum depth 4, subsampling 0.85, column subsampling 0.9, early-stopping rounds 12, and random seed 42. Its estimator settings, rather than the neural epoch, optimizer, and clipping fields in the generic run configuration, determine its training procedure.

F Extended NEXT Classification Results

Table 10: NEXT classification candidates reevaluated with 600 regular 5 keV bins on 0–3000 keV plus one overflow bin above 3000 keV. Params (M) counts trainable neural parameters in millions. A dash denotes an inapplicable parameter count. [†]Architecture instantiated across datasets; no new model is fitted.

Model	Params (M)	Inclusive AUC $\uparrow$	Matched AUC $\uparrow$	I $\uparrow$
<i>Projection and voxel models</i>				
Two-conv CNN	0.0014	0.953280	0.952992	0.966472
Energy-skip CNN	0.0014	0.923394	0.922735	0.967952
Residual spatial CNN	1.2288	0.979030	0.978966	0.969531
Multi-view CNN [†]	0.8077	0.976636	0.976587	0.970993
Multi-scale CNN	0.8659	0.979270	0.979046	0.973145
Dense 3D ResNet	0.6884	0.984115	0.983956	0.971101
Projection MLP mixer	0.5391	0.950352	0.950086	0.972549
Submanifold ResNet	0.2982	0.991574	0.991414	0.967824
<i>Point-set models</i>				
DeepSets	0.0756	0.969660	0.969459	0.972602
PointNet++	0.1645	0.985459	0.985354	0.972546
PointMLP	0.4435	0.994033	0.993814	0.967686
Rigid KPConv	0.3910	0.986654	0.986341	0.969283
<i>Graph models</i>				
Static GINE [†]	0.4799	0.994801	0.994625	0.970265
ParticleNet-lite	0.1505	0.992161	0.991971	0.968044
EGNN	0.6405	0.890742	0.889878	0.968064
GravNet	0.2936	0.991316	0.991070	0.969637
DimeNet-lite	0.1696	0.992939	0.992716	0.968001
<i>Sequence, state-space, topology and hybrid models</i>				
BiGRU [†]	0.7340	0.979563	0.979377	0.971051
Dilated TCN	0.6945	0.973450	0.973183	0.974533
PointMamba-lite [†]	0.3170	0.925742	0.925386	0.971387
Persistence/PersLay	0.0633	0.918258	0.917657	0.968522
Topology XGBoost	—	0.937279	0.936938	0.970396
CNN-GNN	0.3420	0.991417	0.991260	0.970276

G Transformer Representation Details

Table 11 records the concrete dataset-specific implementations of the three tokenization families used in Section 3.2. Here, d_f is the token content width and d_p is the coordinate width. A token count preceded by $\leq$ denotes a maximum sequence length; shorter events are masked and padded only for batching.

NEXT and SuperNEMO use three-dimensional detector position coordinates. MJD uses normalized time together with normalized waveform level and local temporal change. EXO-200 uses normalized time, within-block channel position, detector side, and indicators for the U-wire, V-wire, and APD sensor families.

Table 11: Dataset-specific Transformer tokenization configurations. Internal feature padding, such as the channel padding used by EXO-200 raw patches, does not introduce additional Transformer tokens.

Dataset	Family	Tokens	d_f/d_p	Construction
NEXT	Entity	≤ 512	2/3	Individual detector hits selected under a fixed token cap. Token content describes the local hit measurement, while centered three-dimensional coordinates provide position.
NEXT	Region	≤ 512	2/3	Hits are aggregated within occupied three-dimensional voxels of width 15 detector-coordinate units. If necessary, the most occupied voxels are retained under the token cap.
NEXT	Summary	≤ 512	4/3	Hits are ordered using a three-dimensional Morton ordering and partitioned into balanced spatial groups. Each token summarizes the measurements within one group.
MJD	Entity	500	2/3	The length-3800 waveform is represented by a deterministic mixture of uniformly spaced and high-importance samples. Local context spans nine waveform samples.
MJD	Region	500	20/3	The waveform is divided into 512 non-overlapping patches containing 20 raw amplitudes each.
MJD	Summary	500	2/3	The same 512 temporal regions are represented by their segment mean and root-mean-square amplitude.
EXO-200	Entity	504	2/6	Entities are selected independently within each of six sensor blocks using equal proportions of uniform and waveform-salient samples. Each token stores amplitude and nine-sample local RMS.
EXO-200	Region	504	150/6	Six sensor blocks are divided into seven channel regions and twelve time regions. Each token retains up to 6×25 raw waveform values; smaller channel regions are padded only within the token feature.
EXO-200	Summary	504	4/6	The same sensor-temporal regions are described by mean, RMS, maximum absolute amplitude, and the difference between the second- and first-half means.
SuperNEMO	Entity	≤ 512	4/3	Individual tracker hits form tokens under a deterministic event-level cap, with centered three-dimensional detector coordinates.
SuperNEMO	Region	≤ 512	8/3	Hits are aggregated into occupied cubic spatial regions with width 88 mm. Occupancy-based selection is used if an event exceeds the token cap.
SuperNEMO	Summary	≤ 16	8/3	Hits are Morton ordered and divided into at most sixteen balanced spatial groups, each represented by aggregated event statistics.